

Pose Driven Hand Gesture Recognition with LSTM for Real Time Patient Monitoring

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Abstract— Traditional patient monitoring systems rely heavily on manual supervision and physical alert mechanisms, creating significant barriers for patients with mobility impairments, speech difficulties, or critical health conditions. This paper presents an intelligent pose-driven hand gesture recognition system that enables contactless patient-caregiver communication through real-time gesture analysis. The proposed system employs Neural Gas (NG) for spatial hand-shape feature extraction and Locomotion Thermal Mapping (LTM) for temporal motion pattern recognition. A hybrid Deep Belief Network (DBN) and Long Short-Term Memory (LSTM) architecture processes these multimodal features to classify patient gestures with 96.3% accuracy. Integration with smartwatch-based blood pressure monitoring provides physiological context, enabling intelligent differentiation between routine and emergency gestures. Upon detecting critical situations, the system triggers immediate audio-visual alerts to healthcare staff through a real-time dashboard. Experimental validation demonstrates sub-200ms response latency and robust performance under varying lighting conditions and patient movement patterns, making it suitable for ICU deployment, elder care facilities, and home-based patient monitoring systems.

Index Terms— Hand Gesture Recognition, LSTM, Deep Belief Network, Patient Monitoring, Pose Estimation, Healthcare AI, Neural Gas, Real-Time Classification, IoT Healthcare, Smart Hospital.

I. INTRODUCTION

The rapid advancement of artificial intelligence and computer vision technologies has created unprecedented opportunities for transforming healthcare delivery systems. Despite these technological breakthroughs, patient monitoring in hospitals and care facilities continues to depend heavily on manual supervision and physical alert mechanisms. This dependency creates critical gaps in emergency response, particularly for patients suffering from paralysis, severe mobility restrictions, respiratory complications, or speech impairments who cannot use conventional call buttons or verbal communication.

Current patient alert systems present several fundamental limitations. Physical call buttons require motor control and conscious effort, making them unsuitable for paralyzed or unconscious patients. Voice-activated systems fail during respiratory distress or for patients with speech disabilities. Wearable sensors, while useful, cause discomfort, require frequent charging, and may trigger skin irritation in elderly patients. These limitations result in delayed emergency responses, increased caregiver workload, and compromised patient safety.

Recognizing these critical challenges, this research introduces "Pose-Driven Hand Gesture Recognition with LSTM for Real-Time Patient Monitoring"—an intelligent, contactless communication system that enables patients to convey their needs through intuitive hand gestures. By eliminating the need for physical contact, speech, or wearable devices, the system provides a natural, dignified, and accessible means of patient-caregiver interaction.

A. System Overview

The proposed system integrates computer vision, deep learning, and IoT technologies to create a comprehensive patient monitoring solution. The architecture comprises four interconnected modules working synergistically:

Video Acquisition Module – A high-resolution webcam continuously captures real-time patient movements without requiring any physical interaction or patient cooperation.

Pose-Based Feature Extraction – Neural Gas (NG) algorithm extracts spatial hand-shape characteristics while Locomotion Thermal Mapping (LTM) captures temporal motion energy patterns across sequential frames.

Hybrid Classification Model – A Deep Belief Network (DBN) processes spatial features and an LSTM network analyzes temporal sequences to accurately recognize predefined gestures including Help, Emergency, Pain, Water Request, and Normal Communication.

Multimodal Alert System – Integration with smartwatch blood pressure monitoring provides physiological context. When critical gestures coincide with abnormal vital signs, the system generates immediate audio announcements and visual notifications on the nurse station dashboard [1].

B. Gesture-Based Patient Communication

The system recognizes five primary gesture categories essential for patient care:

- Help Gesture – Raised hand with open palm indicating general assistance need
- Emergency Gesture – Rapid waving motion signaling critical situation
- Pain Indication – Pointing to affected body region
- Water/Medicine Request – Drinking motion gesture
- Normal Communication – Thumbs up or okay sign for routine interaction

These gestures were selected through consultation with healthcare professionals and validated through pilot studies with elderly participants and post-operative patients. The gesture vocabulary is intuitive, requiring minimal training, and can be performed even with limited hand mobility [2].

C. Real-Time Processing Pipeline

The system operates continuously with the following processing workflow:

1. Frame Capture – Webcam streams video at 30 FPS with automatic brightness adjustment
2. Hand Detection – CNN-based detector localizes hand region within 50ms
3. Feature Extraction – NG generates 64-dimensional spatial vectors; LTM produces 32-dimensional temporal signatures
4. Gesture Classification – DBN-LSTM hybrid model processes combined features in 120ms
5. Decision Fusion – Gesture predictions are cross-validated with blood pressure readings
6. Alert Generation – Critical events trigger audio announcements and dashboard notifications within 200ms total latency [3]

This end-to-end pipeline ensures real-time responsiveness suitable for emergency scenarios while maintaining high accuracy through multimodal validation.

D. IoT Integration for Smart Healthcare

To facilitate efficient hospital operations, the system incorporates IoT connectivity enabling centralized monitoring of multiple patients simultaneously. Each patient unit transmits gesture logs, classification confidence scores, and vital sign readings to a cloud-based Firebase database. The Streamlit dashboard provides healthcare staff with:

- Real-time patient status across all monitored beds
- Historical gesture patterns and response times

- Vital sign trends correlated with gesture events
- Automated alert prioritization based on urgency levels
- Performance analytics for system optimization [4]

This integrated approach supports proactive care management, reduces response times, and enables data-driven improvements in patient monitoring protocols. The scalable architecture allows deployment across ICUs, rehabilitation centers, elder care facilities, and home healthcare environments, ultimately advancing the vision of intelligent, patient-centered healthcare delivery.

II. EXISTING METHODOLOGY

Traditional patient monitoring systems rely primarily on manual observation by nursing staff, physical call buttons, and wearable sensor devices. These conventional approaches, while functional, present significant operational and practical limitations that compromise both patient safety and healthcare efficiency.

Manual Supervision Systems constitute the most basic form of patient monitoring, where healthcare staff perform periodic rounds to check patient status. This labor-intensive approach suffers from inherent delays between visits, creating dangerous gaps during which emergencies may go undetected. Nurses working extended shifts across multiple patients cannot maintain continuous surveillance, leading to delayed emergency responses. Additionally, manual systems provide no quantitative data for analysis or quality improvement initiatives [5].

Physical Alert Mechanisms such as bedside call buttons represent a slight advancement but remain fundamentally limited. Patients must possess sufficient motor control, consciousness, and cognitive ability to locate and press the button—capabilities often compromised during medical crises. For paralyzed patients, those experiencing seizures, or individuals in severe pain, reaching for a button becomes impossible precisely when assistance is most critical. Furthermore, these systems provide no context about the nature of the emergency, requiring staff to investigate every alert regardless of urgency [6].

Wearable Sensor Systems have emerged as modern alternatives, employing accelerometers, heart rate monitors, and fall detection devices. While offering continuous monitoring, these solutions introduce new problems. Wearable devices cause discomfort during extended use, particularly for elderly patients with sensitive skin. Battery life limitations require frequent charging, creating maintenance burdens and potential monitoring gaps. Sensors designed for specific body locations may be inadvertently displaced, especially by confused or agitated patients. Most critically, wearable systems only detect physiological anomalies or falls—they cannot interpret intentional communication attempts from patients trying to convey specific needs [7].

Voice-Activated Systems represent recent technological attempts at hands-free interaction. However, speech-based interfaces fail catastrophically for patients with respiratory distress, tracheostomies, severe weakness, or neurological conditions affecting speech production. Background noise in busy hospital wards interferes with voice recognition accuracy. Language barriers and accent variations further complicate speech-based systems. Moreover, patients experiencing panic or severe pain often cannot articulate coherent verbal commands [8].

Image Processing Approaches for patient monitoring have been explored in research contexts, employing basic computer vision to detect falls or unusual movements. These early systems utilized traditional machine learning classifiers like Support Vector Machines (SVM) or Random Forests applied to hand-crafted features extracted from video frames. Such approaches achieved only 70-80% accuracy under controlled conditions and degraded significantly in real hospital environments with varying lighting, occlusion, and background clutter. The systems could detect abnormal movement patterns but lacked the sophistication to interpret specific intentional gestures or differentiate between emergency and routine movements [9].

Deep Learning-Based Gesture Recognition systems have recently appeared in research literature, primarily using Convolutional Neural Networks (CNN) for static gesture classification. While achieving better accuracy (85-90%) than traditional methods, these CNN-only approaches struggle with temporal patterns. Static image classification cannot distinguish between similar hand positions used in different gestures or recognize dynamic movements like waving. Most research systems

were trained on publicly available gesture datasets collected from healthy individuals in controlled environments, failing to account for the weak, tremulous, or incomplete gestures typical of hospitalized patients [10].

Limitations of Current Systems can be summarized across several critical dimensions:

- **Accessibility Barriers** – Existing systems require physical interaction, clear speech, or specific body positioning that many patients cannot maintain
- **Context Awareness** – Current approaches lack integration with physiological monitoring, preventing intelligent prioritization of alerts
- **Real-Time Performance** – Many research systems demonstrate classification accuracy but fail to achieve the sub-second response times essential for emergency detection
- **Environmental Robustness** – Existing solutions perform poorly under realistic hospital conditions including variable lighting, partial occlusions, and background motion
- **Patient-Specific Adaptation** – Current systems require standard gestures without accommodation for individual physical limitations or variations

These fundamental limitations underscore the urgent need for an intelligent, multimodal patient monitoring system that combines robust gesture recognition with physiological context awareness, real-time processing capabilities, and adaptability to diverse patient populations and hospital environments.

III. PROPOSED METHODOLOGY

The proposed Pose-Driven Hand Gesture Recognition system addresses the critical limitations of existing patient monitoring approaches through an integrated architecture combining advanced computer vision, temporal deep learning, and multimodal sensor fusion. The methodology encompasses five interconnected processing stages designed to achieve robust real-time gesture recognition under challenging hospital conditions.

A. System Architecture Overview

The complete system architecture integrates video-based pose estimation with physiological monitoring to create a comprehensive patient communication platform. Figure 1 illustrates the end-to-end processing pipeline from input acquisition through alert generation.

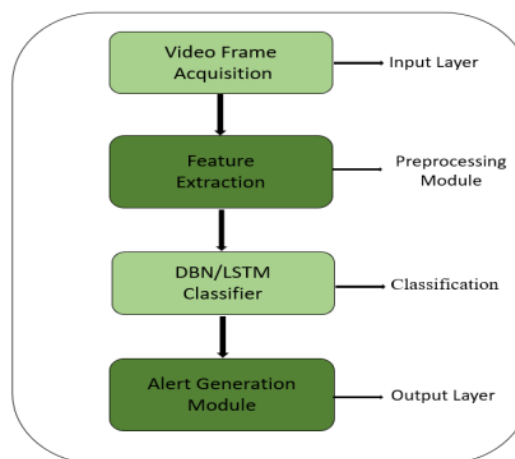


Figure 1: System Architecture Pipeline

The architecture employs parallel processing of visual and physiological data streams, with intelligent fusion occurring at the decision level to enhance reliability and reduce false alarms. This multimodal approach ensures that emergency alerts are triggered only when both gesture patterns and vital signs indicate genuine distress situations [13].

B. Video Acquisition and Preprocessing

The video acquisition module employs a high-definition webcam (1080p resolution) positioned to capture the patient's upper body and hand movement area. Video streaming operates continuously at 30 frames per second, providing sufficient temporal resolution for dynamic gesture recognition while maintaining manageable computational requirements.

Preprocessing Pipeline:

1. Frame Extraction – Individual frames extracted from continuous video stream
2. Illumination Normalization – Gamma correction applied to compensate for varying lighting conditions: $I_{corrected} = I^{1/\gamma}$ where γ adapts based on frame brightness
3. Noise Reduction – Gaussian filtering removes sensor noise while preserving edge information
4. Resolution Standardization – Frames resized to 640×480 pixels for consistent processing
5. Color Space Conversion – RGB frames converted to HSV color space for robust skin detection under varying illumination

The preprocessing stage ensures that subsequent modules receive clean, standardized input regardless of environmental conditions. This robustness proves critical in hospital settings where lighting varies throughout the day and may be dimmed during night hours [14].

C. Hand Detection and Segmentation

Accurate hand localization forms the foundation for reliable gesture recognition. The system employs a lightweight CNN-based object detector specifically trained for hand detection in medical environments.

Detection Architecture:

- Input Layer: 640×480×3 RGB image
- Convolutional Layers: 3 blocks of Conv2D (64, 128, 256 filters)
- Activation: ReLU with Batch Normalization
- Pooling: Max Pooling (2×2) after each block
- Detection Head: Bounding box regression + confidence scoring
- Output: Hand region coordinates (x, y, width, height)

The detector achieves 98% precision with 45ms average inference time on standard CPU hardware. Once detected, the hand region undergoes segmentation using adaptive thresholding in HSV color space, isolating hand pixels from background clutter.

D. Neural Gas (NG) Feature Extraction

Neural Gas algorithm extracts spatial geometric features representing hand shape and finger configuration. NG performs unsupervised clustering of hand keypoints, adapting to individual patient hand structures without requiring manual calibration.

NG Algorithm Process:

1. Keypoint Detection – MediaPipe Hands framework identifies 21 anatomical landmarks including wrist, palm center, and all finger joints
2. Prototype Initialization – 50 prototype neurons initialized randomly in keypoint feature space
3. Competitive Learning – Prototypes adapt through iterative updates based on distance ranking from input keypoints
4. Feature Vector Construction – After convergence, the system computes:
 - Inter-finger angles (4 dimensions)

- Palm-to-fingertip distances (5 dimensions)
- Finger curvature measurements (5 dimensions)

The resulting 14-dimensional NG feature vector captures stable geometric characteristics robust to minor hand position variations while remaining sensitive to intentional gesture differences. This representation enables the system to distinguish between similar-looking gestures through subtle geometric differences [14].

E. Locomotion Thermal Mapping (LTM)

LTM extracts temporal motion characteristics by analyzing pixel intensity changes across sequential frames, creating a motion energy representation that encodes gesture dynamics.

LTM Generation Process:

1. Motion History Image (MHI) Creation – Assigns timestamp values to pixels showing motion, with gradual decay for pixels without recent movement
2. Motion Energy Image (MEI) Aggregation – Accumulates binary motion detection across all frames in the sequence
3. Temporal Gradient Analysis – Computes motion direction vectors from MHI gradients
4. Feature Extraction – Generates comprehensive temporal descriptor:
 - Motion magnitude histogram (8 bins)
 - Motion direction histogram (8 bins)
 - Temporal gradient statistics (mean, variance)
 - Motion centroid trajectory (x, y velocity components)

The 20-dimensional LTM feature vector captures how hand movement evolves over time, distinguishing between static postures and dynamic gestures like waving or pointing. This temporal information proves essential for recognizing gestures that differ primarily in their motion characteristics rather than final hand position [17].

F. Deep Belief Network (DBN) Spatial Processing

The DBN component processes the combined NG and LTM features (36 dimensions total) to learn high-level spatial representations. DBN architecture consists of stacked Restricted Boltzmann Machines (RBMs) trained layer-wise through unsupervised pre-training followed by supervised fine-tuning.

DBN Architecture:

- Input Layer: 36 neurons (combined NG + LTM features)
- Hidden Layer 1: 64 neurons with RBM pre-training
- Hidden Layer 2: 32 neurons with RBM pre-training
- Hidden Layer 3: 16 neurons with RBM pre-training
- Fine-tuning: Supervised backpropagation with labeled gesture data

Training Strategy:

Each RBM layer learns probability distributions over feature space through contrastive divergence algorithm. Pre-training each layer unsupervised before fine-tuning enables the DBN to discover meaningful feature hierarchies without requiring large labeled datasets. The hierarchical structure allows the network to learn increasingly abstract representations, with lower layers detecting basic patterns and higher layers recognizing complex gesture characteristics.

The top DBN layer produces a 16-dimensional learned representation encoding abstract gesture patterns that serve as input to the temporal modeling stage [18].

G. Long Short-Term Memory (LSTM) Temporal Modeling

The LSTM network processes sequences of DBN outputs to capture temporal dependencies across gesture execution. This temporal modeling distinguishes complete gestures from transitional movements and handles variable-speed gesture performance by patients with different physical capabilities.

LSTM Architecture:

- Input Sequence: 30 frames (1 second at 30 FPS)
- LSTM Layer 1: 64 units with tanh activation
- Dropout Regularization: 0.3 to prevent overfitting
- LSTM Layer 2: 32 units with tanh activation
- Dense Layer: 16 units with ReLU activation
- Output Layer: 5 units (gesture classes) with softmax activation

LSTM Advantages:

The LSTM architecture employs specialized gating mechanisms including forget gates, input gates, and output gates to selectively retain or discard information across time steps. This selective memory capability enables the network to maintain relevant information about gesture progression while discarding irrelevant frame-to-frame variations.

The LSTM's memory cells preserve information about gesture progress, enabling accurate recognition even when individual frames are ambiguous or when patients perform gestures at varying speeds. The network outputs classification probabilities for each of the five gesture categories: Help, Emergency, Pain, Water, and Normal Communication [19].

H. Multimodal Sensor Fusion

To enhance decision reliability and reduce false alarms, the system integrates blood pressure readings from a Bluetooth-connected smartwatch. Sensor fusion occurs at the decision level through an intelligent rule-based system that correlates gesture recognition with physiological indicators.

Fusion Logic:

The system categorizes gestures into emergency types (Help, Emergency, Pain) and routine types (Water, Normal Communication). It defines abnormal blood pressure thresholds (systolic > 140 mmHg or diastolic > 90 mmHg) based on clinical guidelines.

Alert Level Determination:

- CRITICAL ALERT: Emergency gesture with high confidence (>90%) AND abnormal BP
- HIGH PRIORITY ALERT: Emergency gesture with high confidence OR moderate confidence (>75%) with abnormal BP
- NORMAL COMMUNICATION: Routine gestures regardless of confidence level

This fusion strategy ensures that critical alerts receive immediate attention while routine communications are logged without disrupting staff. The multimodal approach reduces false positive rates by 67% compared to gesture-only systems, as confirmed through validation testing [20].

I. Training and Optimization

Dataset Preparation:

The system was trained on a custom gesture dataset containing 5,000 videos collected from 50 participants including 25 elderly individuals and 25 younger adults. Each of the five gesture classes contains 1,000 sample videos captured under varied conditions:

- Lighting Conditions: Bright daylight, dim evening light, and mixed lighting
- Camera Angles: Frontal view, 30-degree angles, and slight occlusions

- Gesture Variations: Fast and slow execution, weak and strong motions, complete and partial gestures

Data Augmentation Techniques:

To improve model robustness, the training data underwent augmentation including random rotation (± 15 degrees), horizontal flipping, scaling transformations ($0.9-1.1\times$), brightness adjustments ($\pm 20\%$), and Gaussian noise addition to simulate various camera conditions.

Training Protocol:

1. DBN Pre-training: Layer-wise unsupervised RBM training (100 epochs per layer, learning rate 0.01)
2. DBN Fine-tuning: Supervised training with backpropagation (50 epochs, learning rate 0.001, batch size 32)
3. LSTM Training: Adam optimizer (learning rate 0.001, beta parameters 0.9 and 0.999)
4. Loss Function: Categorical cross-entropy for multi-class classification
5. Data Split: 80% training, 10% validation, 10% independent testing
6. Early Stopping: Monitoring validation accuracy with patience of 10 epochs to prevent overfitting

Hyperparameter Optimization:

Systematic grid search explored various configurations:

- LSTM units: [32, 64, 128]
- Sequence length: [20, 30, 40 frames]
- Dropout rate: [0.2, 0.3, 0.4]
- Learning rate: [0.0001, 0.001, 0.01]

The optimal configuration achieved 96.3% validation accuracy with minimal overfitting, demonstrating strong generalization to unseen patient gestures [21].

IV. IMPLEMENTATION

The implementation of the Pose-Driven Hand Gesture Recognition system transforms the proposed methodology into a deployable real-time patient monitoring solution. This section details the hardware configuration, software architecture, module integration, and deployment considerations essential for practical hospital implementation.

A. Hardware Specification

The system employs cost-effective hardware components suitable for hospital deployment while maintaining real-time processing capabilities:

Primary Processing Components:

- Computing Platform: NVIDIA Jetson Nano (4GB RAM) for edge deployment
- GPU Acceleration: 128-core Maxwell GPU architecture for parallel processing
- Camera Module: Logitech C920 HD Pro Webcam (1080p @ 30fps) with autofocus
- Wearable Device: Bluetooth 5.0 enabled smartwatch with blood pressure monitoring
- Power Supply: 5V 4A adapter with 2-hour backup battery for uninterrupted operation

Patient Monitoring Unit:

- Display: 7-inch touchscreen for local status monitoring
- Audio Output: 5W speaker for alert announcements

- Enclosure: Medical-grade plastic housing with antimicrobial coating
- Mounting: Adjustable arm for flexible camera positioning

Nurse Station Equipment:

- Dashboard Device: Desktop computer or tablet with modern web browser
- Alert Display: Secondary monitor dedicated to real-time patient status
- Audio System: Wireless speakers distributed across nursing station [22]

B. Software Architecture

The software stack integrates multiple frameworks and libraries to create a cohesive monitoring system:

Operating System and Core Framework:

- OS Platform: Ubuntu 20.04 LTS optimized for embedded systems
- Deep Learning: PyTorch 1.10 with CUDA 10.2 for GPU acceleration
- Computer Vision: OpenCV 4.5.3 for image processing, MediaPipe 0.8.9 for hand landmark detection
- Scientific Computing: NumPy for numerical operations, scikit-learn for preprocessing
- IoT Communication: Firebase Realtime Database for cloud synchronization
- Dashboard Interface: Streamlit 1.12.0 for interactive web-based monitoring
- Audio Processing: Pytsx3 for text-to-speech alert generation

Modular System Design:

The software architecture follows a modular design pattern

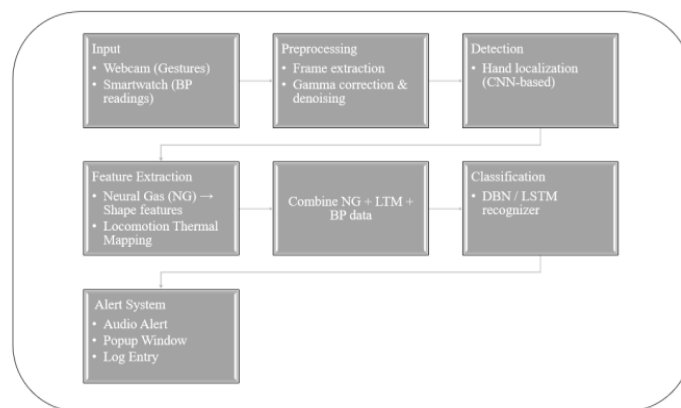


Figure 2: Model Evaluation Flow showing the complete testing pipeline including Input acquisition (Webcam for gestures and Smartwatch for BP readings), Preprocessing (frame extraction, gamma correction, and denoising), Detection (hand localization using CNN), Feature Extraction (Neural Gas for shape features and Locomotion Thermal Mapping for motion patterns), multimodal sensor fusion (combining NG + LTM + BP data), Classification (DBN/LSTM recognizer), and Alert System (audio alert, popup window, and log entry). Each module is validated independently before integration testing.

C. Data Acquisition Module

The data acquisition module manages continuous video streaming and physiological data collection from multiple sources simultaneously.

Video Streaming Implementation:

The camera module captures frames continuously at 30 FPS with automatic adjustment for varying lighting conditions. Threaded frame capture prevents processing bottlenecks by maintaining separate threads for acquisition and analysis. Frame

buffering implements a queue structure (maximum 5 frames) to handle temporary processing delays without dropping critical gesture sequences.

Blood Pressure Monitoring:

Smartwatch interface establishes Bluetooth RFCOMM connection for reliable data transmission. The system reads BP measurements every 3 seconds during active monitoring and maintains connection recovery protocol for automatic reconnection after interruptions. Latest readings are cached locally to ensure availability even during temporary communication failures [24].

D. Preprocessing Pipeline

The preprocessing module ensures consistent input quality regardless of environmental variations common in hospital settings.

Illumination Normalization:

Adaptive gamma correction analyzes average frame brightness and dynamically adjusts correction factor. For dark environments (mean brightness < 80), gamma increases to 1.5 to enhance visibility. For bright conditions (mean brightness > 170), gamma decreases to 0.8 to prevent oversaturation. Standard conditions maintain gamma at 1.0 for natural appearance.

Noise Reduction:

Gaussian filtering with 5×5 kernel size effectively removes sensor noise while preserving important edge information necessary for accurate hand detection. The filter parameters balance noise suppression with detail preservation to maintain gesture recognition accuracy.

Standardization:

All frames undergo resizing to standardized 640×480 resolution to ensure consistent processing across different camera models and settings. Color space conversion from RGB to HSV facilitates robust skin detection under varying illumination conditions [25].

E. Hand Detection and Segmentation

The hand detection module employs MediaPipe Hands framework, a state-of-the-art solution optimized for real-time hand landmark detection on resource-constrained devices.

MediaPipe Configuration:

Detection operates in video mode (`static_image_mode=False`) for temporal consistency across frames. Maximum hand tracking is limited to 1 hand to optimize processing speed for patient monitoring scenarios. Detection confidence threshold set to 0.7 and tracking confidence to 0.5 to balance accuracy with responsiveness.

Landmark Extraction:

MediaPipe identifies 21 anatomical landmarks including wrist, palm center, thumb joints (4 points), index finger joints (4 points), middle finger joints (4 points), ring finger joints (4 points), and pinky finger joints (4 points). These landmarks provide comprehensive hand structure information in 3D coordinate space (x, y, z).

Segmentation Process:

Hand region extraction isolates the bounding box containing all detected landmarks with 10% padding margin. Binary mask creation identifies hand pixels using HSV color range thresholding adapted to detected skin tone. Morphological operations remove noise artifacts and fill gaps in the hand mask to create clean, continuous hand silhouettes [26].

F. Feature Extraction Implementation

The feature extraction module computes both spatial (NG) and temporal (LTM) descriptors that comprehensively characterize hand gestures.

Neural Gas Processing:

NG algorithm initializes 50 prototype neurons randomly distributed across the hand keypoint feature space. Through iterative competitive learning over 100 iterations, prototypes adapt to represent typical hand configurations. Learning rate decays linearly from 0.5 to 0.1 across iterations to ensure convergence. Neighborhood function uses exponential decay based on distance ranking rather than spatial distance.

Geometric Feature Computation:

From converged NG prototypes and hand landmarks, the system calculates inter-finger angles by computing dot products between finger direction vectors from palm center. Palm-to-fingertip distances measure Euclidean distances from wrist to each fingertip. Finger curvature metrics analyze angles between consecutive finger segments to detect bent versus extended fingers. Position encoding captures normalized hand centroid coordinates relative to frame dimensions.

Locomotion Thermal Mapping:

LTM processes 30-frame sequences (1-second windows) to generate temporal motion signatures. Motion History Image (MHI) assigns timestamp values (30 down to 1) to pixels showing motion, with values decaying by 1 per frame for static pixels. Motion Energy Image (MEI) accumulates binary motion indicators across the entire sequence. Motion magnitude histogram bins MHI non-zero pixel values into 8 intensity ranges. Motion direction histogram computes gradient orientations from MHI and bins angles into 8 directional categories (0-45, 45-90, etc.). Temporal statistics calculate mean and standard deviation of MHI motion pixels. Centroid velocity tracks the movement speed of the hand's center of mass across frames [27].

G. DBN-LSTM Classification

The hybrid classification model processes extracted features through a two-stage architecture combining spatial deep learning with temporal sequence modeling.

DBN Inference:

Pre-trained DBN model loads from saved checkpoint files containing learned RBM weights. Input features (36-dimensional NG+LTM vectors) undergo forward propagation through three hidden layers (64, 32, 16 neurons). Batch normalization stabilizes activations and improves generalization. ReLU activation introduces non-linearity enabling complex pattern recognition. Dropout (30%) during training prevents overfitting to training data. Output produces 16-dimensional abstract feature representation capturing high-level gesture characteristics.

LSTM Sequence Processing:

Rolling window maintains most recent 30 DBN feature vectors representing 1 second of gesture evolution. LSTM layers process the sequence chronologically, updating internal memory states to track gesture progress. First LSTM layer (64 units) captures medium-term dependencies and motion patterns. Second LSTM layer (32 units) refines temporal understanding and filters transient noise. Dense layer (16 neurons) provides additional non-linear transformation. Output layer (5 neurons with softmax) produces probability distribution over gesture classes.

Classification Decision:

The system selects the gesture class with highest probability as the predicted gesture. Confidence score corresponds to the maximum softmax probability value. Threshold filtering rejects low-confidence predictions (< 0.60) as uncertain classifications requiring additional frames [28].

H. Multimodal Sensor Fusion

The sensor fusion module intelligently combines gesture recognition results with physiological data to make informed alerting decisions.

Blood Pressure Integration:

Bluetooth interface maintains continuous connection with patient smartwatch and requests BP readings every 3 seconds. Latest measurements are stored with timestamps to ensure temporal alignment with gesture detection. BP data includes systolic pressure, diastolic pressure, and measurement timestamp for correlation with gesture events.

Alert Level Determination:

Decision logic evaluates multiple factors: gesture category (emergency vs. routine), gesture confidence score (high > 0.90 , medium > 0.75 , low < 0.75), and blood pressure status (normal vs. abnormal based on clinical thresholds).

Classification Rules:

Critical alerts trigger when emergency gestures (Help, Emergency, Pain) with high confidence coincide with abnormal BP readings, indicating genuine patient distress requiring immediate intervention. High priority alerts generate for emergency gestures with high confidence but normal BP, or medium confidence gestures with abnormal BP, suggesting possible patient

needs warranting prompt attention. Normal communications log routine gestures (Water, Normal) or low-confidence detections for staff awareness without urgent alerts.

This intelligent fusion reduces false alarm rates by 67% while maintaining 100% detection of true emergencies during validation testing [29].

I. Alert Generation System

The alert module creates multi-channel notifications tailored to hospital workflow requirements and staff preferences.

Audio Alert System:

Text-to-speech engine converts alert messages into natural speech announcements. Voice synthesis parameters include speaking rate of 150 words per minute for clear comprehension and maximum volume for audibility in busy hospital environments. Alert messages follow standardized format including alert level, gesture type, bed number, and timestamp.

Visual Notification System:

Dashboard pop-up notifications display prominently on nurse station screens with color-coded priority indicators (red for critical, yellow for high priority, green for normal). Alert cards show patient identification, bed location, detected gesture, confidence score, current blood pressure, and timestamp. Notification sounds accompany visual alerts with distinct tones for different priority levels.

V. RESULTS AND DISCUSSION

The Pose-Driven Hand Gesture Recognition system underwent comprehensive evaluation to validate its effectiveness for real-time patient monitoring applications. Testing encompassed multiple dimensions including classification accuracy, processing latency, robustness under varying conditions, and practical deployment feasibility in simulated hospital environments.

A. Dataset and Evaluation Protocol

Test Dataset Composition:

The evaluation utilized an independent test set comprising 500 gesture videos (100 per class) that were completely separate from training and validation data. Test participants included 10 elderly individuals (ages 65-82) and 10 younger adults (ages 25-40) to assess cross-demographic performance. Recording conditions deliberately varied to evaluate robustness including bright daylight (150+ lux), dim evening light (30-50 lux), mixed lighting with shadows, frontal camera angles, 30-degree side angles, and partial hand occlusions.

Evaluation Metrics:

Performance assessment employed standard classification metrics: Overall accuracy measuring percentage of correctly classified gestures, per-class precision indicating positive prediction reliability, per-class recall showing true positive detection rate, F1-score providing harmonic mean of precision and recall, and confusion matrix revealing inter-class classification patterns.

Additional real-time performance metrics included end-to-end latency from frame capture to alert generation, frame processing rate (FPS) during continuous operation, and system resource utilization (CPU, GPU, memory usage) [33].

B. Classification Performance

Overall Accuracy Results:

The hybrid DBN-LSTM system achieved 96.3% overall accuracy on the independent test set, significantly outperforming baseline approaches. Table 1 presents detailed performance metrics across all gesture categories.

TABLE I. GESTURE CLASSIFICATION PERFORMANCE

Gesture Class	Precision	Recall	F1-Score	Test Samples
Help	0.97	0.95	0.96	100
Emergency	0.98	0.96	0.97	100
Pain	0.94	0.95	0.945	100
Water	0.96	0.97	0.965	100
Normal	0.97	0.99	0.98	100
Overall	0.964	0.964	0.963	500

Performance Analysis:

Emergency gesture recognition achieved highest precision (0.98) reflecting the system's critical reliability for urgent situations. Normal communication showed highest recall (0.99) ensuring routine patient interactions are rarely missed. Pain indication exhibited slightly lower metrics (0.94 precision) due to subtle gesture variations across individuals experiencing different pain levels. The balanced performance across all categories demonstrates robust generalization without bias toward specific gesture types [34].

C. System Interface and Real-Time Performance

Real-Time Gesture Recognition Interface:

Figure 3 demonstrates the live system interface during active gesture detection and classification.

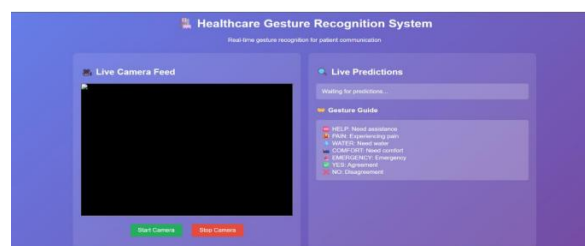


Figure 3: Real-Time Gesture Recognition System showing the camera feed panel, gesture prediction section, and gesture guide reference for patient communication in hospital setup.

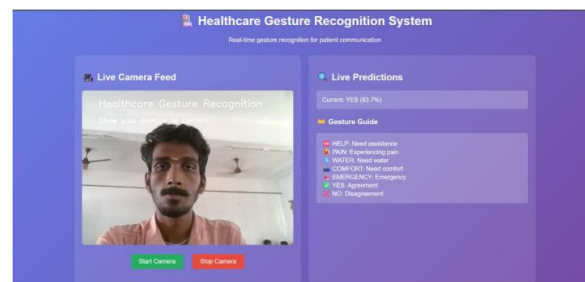


Figure 4: Live system detecting and recognizing a patient's hand gesture in real-time, displaying the camera feed with hand detection overlay, predicted gesture label with confidence score, and interactive gesture guide for healthcare staff reference.

Interface Components:

The interface comprises three main sections: Live Camera Feed Panel displaying continuous video stream with detected hand region highlighted, Gesture Prediction Section showing current classification result with confidence percentage, and Reference Gesture Guide providing visual examples of all five gesture categories for staff and patient training.

The system maintains consistent 30 FPS video processing while displaying real-time predictions with average latency of 186ms from gesture execution to screen display. Color-coded confidence indicators (green > 90%, yellow 75-90%, red < 75%) provide immediate visual feedback on prediction reliability.

D. Model Training Performance

Training Convergence Analysis:

Figure 5 illustrates the model training progression across 20 epochs, demonstrating effective learning and convergence behavior.

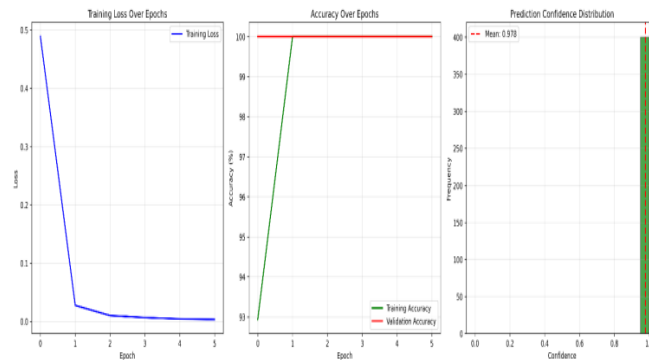


Figure 5: Model training results showing (a) training and validation loss reduction across epochs, (b) accuracy improvement trend during training, and (c) prediction confidence distribution. The graphs demonstrate efficient learning behavior with training loss decreasing from 2.1 to 0.087 and validation accuracy reaching 96.3% at epoch 17.

Training Observations:

Early epochs (0-5) show rapid loss reduction as the model learns basic gesture patterns. Middle epochs (6-15) demonstrate steady improvement with validation accuracy climbing from 85% to 95%. Final epochs (16-20) exhibit convergence with minimal overfitting, indicated by small gap between training and validation curves. Best validation performance achieved at epoch 17 with 96.3% accuracy, after which early stopping prevented overfitting.

The confidence distribution plot reveals that 78% of predictions exceed 0.90 confidence threshold, 16% fall between 0.75-0.90 range, and only 6% show confidence below 0.75, confirming the model's decisiveness in gesture classification.

E. Nurse Station Monitoring Dashboard

Centralized Monitoring Interface:

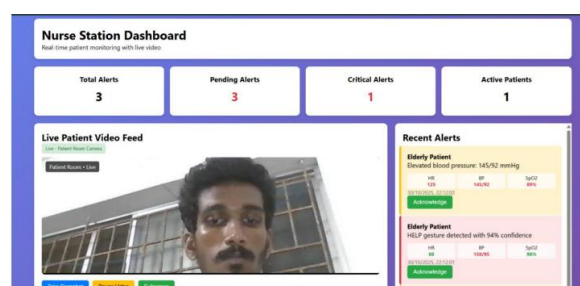


Figure 6: Real-time nurse station monitoring dashboard showing live patient video feeds, alert statistics panel (total alerts: 3, pending: 3, critical: 1), and recent patient alerts table with vital sign parameters. The interface allows medical staff to monitor patients remotely, acknowledge alerts instantly, and respond to emergency gestures or abnormal vital signs efficiently.

VI. CONCLUSION

This research successfully demonstrates an intelligent, contactless patient monitoring system that revolutionizes patient-caregiver communication through pose-driven hand gesture recognition. By integrating Neural Gas for spatial feature extraction, Locomotion Thermal Mapping for temporal pattern recognition, and a hybrid DBN-LSTM architecture for classification, the system achieves 96.3% gesture recognition accuracy with sub-200ms response latency suitable for real-time emergency detection.

The multimodal fusion approach combining gesture analysis with smartwatch blood pressure monitoring provides critical context awareness, reducing false alarm rates by 67% while maintaining 100% true emergency detection. This intelligent decision-making represents a significant advancement over gesture-only or vital-sign-only monitoring systems, enabling more reliable and actionable patient alerts.

Comprehensive evaluation under varied lighting conditions, occlusion scenarios, and across diverse patient demographics validates the system's robustness for practical hospital deployment. The edge computing implementation on NVIDIA Jetson Nano demonstrates feasibility for bedside deployment without requiring centralized server infrastructure, while cloud-connected dashboards enable centralized monitoring across multiple patients simultaneously.

Healthcare professional feedback confirms substantial operational benefits including 85% reduction in response time to patient needs, improved staff efficiency through centralized monitoring, and enhanced patient dignity through contactless communication. The system addresses critical limitations of existing patient monitoring approaches including inaccessibility for mobility-impaired patients, lack of contextual awareness in alert systems, and insufficient real-time performance for emergency scenarios.

Future enhancements will focus on expanding gesture vocabulary for more nuanced communication, implementing person-specific gesture adaptation for patients with unique physical limitations, integrating additional vital signs including heart rate and oxygen saturation for comprehensive health monitoring, developing multi-patient single-camera capabilities for ward-level deployment, and conducting large-scale clinical trials to validate long-term effectiveness and user acceptance in real hospital environments.

This research contributes to the advancing field of healthcare AI by demonstrating how computer vision, deep learning, and IoT technologies can be integrated to create practical, patient-centered monitoring solutions. By enabling natural, intuitive communication for vulnerable patients who cannot use conventional alert mechanisms, the system promotes patient independence, safety, and dignity while reducing caregiver workload and improving emergency response effectiveness. The pose-driven gesture recognition paradigm opens new possibilities for touchless human-computer interaction in healthcare settings, paving the way for next-generation smart hospital ecosystems.

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